

Sprint 2 Plan

SlugRoute

{Week 6: 5/05/26 11:59pm}

Revision 2 {4/21/2026}

Burn Down Chart

 **Sprint 2 Burn Down Chart**

Scrum Board

 **Sprint 2 Scrum Board**

Screenshots

 **SlugRoute Screenshots**

Goal:

Fully integrate GoogleMaps API, and have a mobile-first, friendly UI.

User Stories

1. As a student, I want to see my classes placed accurately on a map of UCSC with restricted boundaries and interactive popups. 5

1. Configure the Google Maps JS SDK to initialize with a custom mapId. Implement LatLngBounds to restrict the user's view and panning to the UCSC campus perimeters. (4)
2. Develop the handshake between the Go API and the frontend that maps building names from the database to their corresponding Latitude/Longitude coordinates. (5)
3. Create a robust engine that clears and redeploys markers based on search results, ensuring Lat/Lng data is converted into interactive map pins in real-time. (5)
4. Engineer the "Info Box" logic that triggers on pin-clicks. This includes formatting the HTML content to display room numbers, lecture times, and instructors, and ensuring the popup is responsive and "filter-aware" (hides meetings if they are filtered out). (6)

2. As a user, I want a fast and responsive interface so that I can interact with my course data in real-time, anywhere, without the entire page having to reload. 13

1. Rewrite the core layout in style.css so that mobile is the default. Use media queries to apply the 350px sidebar only on screens wider than 768px. (8)
2. Update script.js so that when the sidebar is toggled on mobile, it slides over the map (z-index) rather than pushing the map container. (2)

3. Update CSS to ensure all interactive elements (search buttons, course cards, filter checkboxes) have a minimum touch target of 44x44px. (1)
4. Wrap the search input in a <form> and implement event.preventDefault() to allow mobile Enter keys to search without refreshing the page. (1)
5. Create a CSS animation for loading cards that appear in the sidebar while the Go API is fetching results. (2)

3. As a student, I want to search and see colored pins on a building to distinguish between different classes and meeting types. 5

1. Replace the current color array with a hand-picked set of 20 high-contrast colors to ensure markers are distinguishable. (1)
2. Audit and refactor the ColorManager to ensure the same hex code is used for the sidebar border, the map marker, and the InfoWindow header for a specific class. (2)
3. Implement crisp SVG paths for ★ (LEC), ■ (LAB), and ▲ (DIS) within the createMarkerElement function for the Google Maps Advanced Marker API. (4)
4. Refactor the clustering logic so that if multiple meeting types occur in one building, the "Group" (Circle) icon is rendered with the number of meetings in the middle. (8)
5. Add JS listeners to markers so that hovering over a pin triggers a "bounce" or "scale" animation on the corresponding sidebar card. (4)

4. As a student, I want to add all my classes to one view so that I can see where they all take place. 8

1. Ensure the searchCourse function appends new results to the global state rather than overwriting previous searches. (2)
2. Implement map.fitBounds() logic that triggers whenever a new class is added, ensuring the zoom level fits all active pins. (3)
3. Create an "Add All Saved" button in the sidebar that iterates through localStorage and pushes all saved items into the active map view. (4)
4. Update the sidebar rendering to display primary lecture times (e.g., "MWF 10:40AM - 11:45AM") directly on the course cards for quick reference. (3)

5. As a student, I would like to have full control over the visibility of each meeting type and the ability to manage my saved list. 5

1. Add visibility control icons to individual course cards, allowing users to hide or show specific sections on the map without removing the data from their current results. (4)
2. Refine the checkbox filtering system to perform high-performance batch updates on marker visibility (LEC vs. DIS vs. LAB) across the entire map instance. (3)
3. Ensure the toggleSaveCourse function maintains absolute parity between the UI state and localStorage, ensuring the "Saved for Later" list persists through page reloads. (2)
4. Update the InfoWindow generation engine to respect active filters; if "Labs" are globally hidden, they should automatically be omitted from the text content of the building popups. (3)

Total Hours: 77

Team Roles

jd Cody@ucsc.edu :	<i>Developer</i>
jlores@ucsc.edu :	<i>Developer, Product Owner</i>
fveloz@ucsc.edu :	<i>Developer</i>
avalidesg@ucsc.edu :	<i>Developer, Scrum Master</i>

Initial task assignment

jd Cody@ucsc.edu :	<i>User story 1, task 4</i>
jlores@ucsc.edu :	<i>User story 1, task 1</i>
fveloz@ucsc.edu :	<i>User story 1, task 3</i>
avalidesg@ucsc.edu :	<i>User story 1, task 2</i>

Scrum times

TA meeting: Monday (10am-11am)
Team meeting 1: Tuesday (5pm-7pm)
Team meeting 2: Thursday (5pm-7pm)
Team meeting 3: Sunday (12pm-2pm)